

Jane Liu

Vancouver, BC | Canadian Permanent Resident | xiangliu2011@gmail.com

SUMMARY

10+ years of professional software development experience in web and mobile applications across **Banking, Healthcare, Retail, Education** and **Manufacturing domains**, with a strong focus on scalability, reliability, and clean architecture.

Expertise in **HTML5, Tailwind CSS, JavaScript, TypeScript, React, React Native, Redux, Next.js, Node.js, Express, Java, Spring Boot, and Python (Flask/FastAPI), Django**.

Strong focus on Accessibility (**A11y**) best practices, implementing **ARIA** and **WCAG** standards to deliver high-performance, inclusive, and user-friendly applications.

Experienced in **microservices, RESTful and GraphQL APIs**, and **distributed systems** leveraging **Kafka**.

Skilled in **scalable backend architecture, cloud platforms (AWS, GCP)**, and **containerization (Docker, Kubernetes)**.

Proficient in **CI/CD pipelines** using **Jenkins and GitHub Actions**.

Experienced with relational databases (**MySQL, PostgreSQL, SQLite**) and **NoSQL** databases (**DynamoDB, MongoDB**), as well as ORMs (Sequelize, SQLAlchemy), including query optimization, data modeling, and API integration.

Advocate of **Test-Driven Development (TDD)**, with hands-on experience in **unit testing (Jest, Mocha, React Testing Library, JUnit)** and **end-to-end testing (Cypress, Playwright)**.

Experienced in utilizing **MCP Server** to access Figma and automatically generate reusable UI components, streamlining the front-end development process.

Integrated **LLMs** (OpenAI GPT-4, Hugging Face transformers) into financial knowledge retrieval workflows using **LangChain** and vector databases (Pinecone, FAISS), enabling **RAG** for capital offers search.

Leveraged **GitHub Copilot, Cursor, WindSurf, Tabnine, and Kite** to accelerate software development, improve code quality, and reduce repetitive coding tasks.

Expertise in **Agile methodologies (SCRUM frameworks)**, with proficiency in **JIRA** for issue management and bug tracking.

PROFESSIONAL EXPERIENCE

Walmart Global Tech

May 2022 - Sep 2025

Senior Software Engineer

Sunnyvale, CA

- Enhanced a seller-facing payment and capital offers platform serving the U.S., Mexico, and Canada, delivering scalable solutions for 10M+ daily transactions.
- Developed scalable **React/Next.js (TypeScript)**, **TanStack Query** applications with advanced state management (**Redux, Context**), improving page performance by 30% and user engagement by 20%.
- Designed and implemented **Python-based APIs** using **Flask and FastAPI**, integrating with ReactJS frontend and deployed on AWS ECS with Docker/Kubernetes orchestration.
- Implemented server-side rendering (**SSR**) and static site generation (**SSG**) in **Next.js**, reducing initial load times by 40% and boosting SEO rankings.
- Designed and deployed modular microservices using **Node.js (TypeScript, Express/NestJS)**, **Java (Spring Boot)**, **Python**, with **JWT OAuth2 security, Redis, Kafka, GraphQL APIs**, improving system reliability and maintainability.
- Implemented serverless **microservices** by **AWS Lambda** and **gateway**, orchestrated **EKS** to ensure 99.9% SLA uptime.
- Collaborated with design, product, and AI/ML teams to bring smart features from concept to production, accelerating delivery using AI-assisted tools (**Copilot, Cursor, WindSurf**) and **AWS Bedrock**.
- Conducted functional testing using **Jest, JUnit**, and contributed to CI/CD pipelines via **Playwright, UI Shield**, ensuring high code quality.
- Utilized **Hibernate ORM** with **MySQL** to store structured data such as user credentials, roles, and account settings, and imported **MongoDB** to handle unstructured data like the user activity logs and related images.
- Utilized **Apache Kafka** to handle real-time event streaming in the payment commission platform, enabling high scalability and fault-tolerant processing during peak traffic periods.

Filamento

Feb 2021 - Apr 2022

Senior Software Engineer

Sunnyvale, CA

- Delivered responsive **React/Next.js (TypeScript)** and **React Native** UIs, improving cross-platform performance and usability.
- Developed efficient and highly scalable **REST** based web services to build E-commerce solutions using the backend framework of **Spring MVC**, and integrated external partner APIs to scale up the business.
- Developed OMS microservices using **Java (Spring Boot)** and **GraphQL APIs**, enhancing order tracking, quotes, and fulfillment workflows, reducing order processing errors by 20%.
- Built WMS with **Java, Kafka, Redis**, optimizing inventory tracking and picking/packing processes, reducing fulfillment time by

30% across regional warehouses.

- Created TMS features using **Python (FastAPI)** and mapping APIs, reducing transportation costs by 15% and improving delivery reliability.
- Conducted functional testing using **Jest, JUnit**, and contributed to CI/CD pipelines via **Playwright, UI Shield**, ensuring high code quality.

Integem

Apr 2015 - Aug 2020

Software Engineer

Cupertino, CA

- Designed and delivered an AR-based immersive learning platform for 50K+ students using **React, React Native (TypeScript), Redux, Python, Go**, and created interactive STEM/game modules.
- Created prototypes with Figma/XD and built cross-browser responsive UI with SASS/Bootstrap, reducing rendering times by 15%.
- Built backend microservices with **Java (Spring Boot)** to manage user accounts, course content, and progress tracking, ensuring seamless integration with frontend via **REST APIs**.
- Implemented **RBAC** with AccessControl and **JWT**-based **OAuth2** via **Passport.js (Azure AD, Auth0)**; developed secure API gateways with **CORS**, throttling, and real-time updates via **Socket.io** and **GraphQL** resolvers for cross-service data streaming.
- Implemented the UI using **Material UI**, focusing on cross-browser compatibility and user-friendly functionalities.
- Developed AWS-based infrastructure to support scalable web and mobile apps, including **Lambda** for serverless functions, API Gateway for API management, **S3** for content storage, CloudFront for global delivery, RDS (**PostgreSQL/MySQL**) for relational data, and **DynamoDB** for fast key-value access.
- Conducted end-to-end testing with **Cypress, Jest, and Mocha**, achieving 85%+ code coverage and reducing production issues by 20%.
- Developed 3D assets and interactive experiences with **Maya and Blender** (modeling, texturing, rigging, animation, and rendering), integrated into **Unity and Unreal Engine** pipelines.

EDUCATION

Academy of Art University

Master, Web Development

Feb 2010 - May 2013

San Francisco, CA